Sisteme de ecuații diferențiale

$X' = AX$ $\quad X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ $\quad x_1, \dots, x_n : \mathbb{R} \longrightarrow \mathbb{R}$ derivabile

t variabila

$A \in \mathcal{M}_n(\mathbb{R})$

$n = 2$ $\quad \begin{cases} x' = ax + by \\ y' = cx + dy \end{cases}$ $\quad X = \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$P_A(X) = (X - \lambda)^2$ $\quad a(\lambda) = 2$ $\quad g(\lambda) = 1$ $\quad (= \dim_{\mathbb{R}} V_\lambda)$

$\Rightarrow$ sistem fundamental de soluții $\quad X_1(t) = e^{\lambda t} v_1$

$X_2(t) = e^{\lambda t}(t v_1 + v_2)$ unde

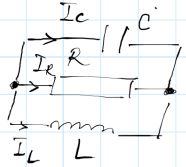
$v_2 \in \mathbb{R}^2$, $v_2 \notin V_1$

$v_1 = (A - \lambda I_2) v_2 \in V_1$

$\Rightarrow$ soluția generală $\quad X(t) = C_1 X_1(t) + C_2 X_2(t)$

$= C_1 e^{\lambda t} v_1 + C_2 e^{\lambda t}(t v_1 + v_2)$

Exemplu



$$\begin{cases} I_C + I_R + I_L = 0 \\ U_C = U_L = U_R \end{cases}$$

rezistor : $I_R(t) = \dfrac{U_R(t)}{R}$

condensator : $I(t) = \dfrac{\Delta q(t)}{\Delta t} = C \cdot U_C'(t)$ $\quad q(t) = C \cdot U_C(t)$

bobină : $U_L(t) = L \cdot I_L'(t)$

$$\begin{cases} I_L'(t) = \dfrac{1}{L} U_L(t) \\ U_L'(t) = \dfrac{1}{C}\left(-I_L(t) - \dfrac{U_L(t)}{R}\right) \end{cases}$$

$x(t) = I_L(t)$

$y(t) = U_L(t)$

$$\Rightarrow \begin{cases} x' = \dfrac{1}{L} y \\ y' = -\dfrac{1}{C} x - \dfrac{1}{RC} y \end{cases}$$

$R = 1\,\Omega$, $C = 1\,F$, $L = 4\,H$

$x(0) = 1\,A$, $y(0) = 2\,V$

$$\begin{cases} x' = \frac{1}{4} y \\ y' = -x - y \end{cases} \qquad x(0) = 1 \qquad y(0) = 2$$

$$A = \begin{pmatrix} 0 & 1/4 \\ -1 & -1 \end{pmatrix}$$

$$P_A(\lambda) = \lambda^2 + \lambda + \frac{1}{4} = \left(\lambda + \frac{1}{2}\right)^2 \qquad \lambda = -\frac{1}{2}$$

$$V_\lambda : \begin{cases} \frac{1}{4} y = -\frac{1}{2} x \\ -x - y = -\frac{1}{2} y \end{cases} \Rightarrow y = -2x$$

$$V_\lambda = \left\{ \begin{pmatrix} x \\ -2x \end{pmatrix} \mid x \in \mathbb{R} \right\}$$

$v_2 \in \mathbb{R}^2 \quad v_2 \notin V_\lambda \qquad v_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$$v_1 = (A - \lambda I_2) v_2 = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ -1 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\Rightarrow \quad X_1(t) = e^{\lambda t} v_1 = e^{-t/2} \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad X_2(t) = e^{\lambda t} (t v_1 + v_2) = e^{-t/2} \left(t \begin{pmatrix} 1 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right)$$

$$= e^{-t/2} \begin{pmatrix} t+2 \\ -2t \end{pmatrix}$$

$\Rightarrow$ sol. generale $\quad X(t) = C_1 e^{-t/2} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + C_2 e^{-t/2} \begin{pmatrix} t+2 \\ -2t \end{pmatrix}, \quad C_1, C_2 \in \mathbb{R}$

$$\lim_{t \to \infty} X(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \forall C_1, C_2$$

$$X(0) = \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \Rightarrow \begin{matrix} C_1 + 2C_2 = 1 \\ -2C_1 = 2 \end{matrix} \qquad \begin{matrix} C_2 = 1 \\ C_1 = -1 \end{matrix}$$

$$x(t) = -e^{-t/2} + e^{-t/2}(t+2) = e^{-t/2}(t+1) \searrow 0$$

$$y(t) = 2e^{-t/2} - 2te^{-t/2} = 2e^{-t/2}(1-t) \nearrow 0$$

$X' = AX$ $\quad$ A a valori propri complessi coniugati.

$P_A(x) = 0$ $\quad$ $\lambda = \alpha + i\beta$, $\beta \neq 0 \Rightarrow V_\lambda = Sp\{v\}$, $v \in \mathbb{C}^2$

$\bar{\lambda} = \alpha - i\beta \Rightarrow V_{\bar{\lambda}} = Sp\{\bar{v}\}$

$(Av = \lambda v \Rightarrow A\bar{v} = \bar{\lambda}\bar{v})$

$\Rightarrow Z(t) = e^{\lambda t} v$ soluzione complessa a sist. di ec. diff. $X' = AX$

$X_1(t) = Re\, Z(t)$ $\quad$ $X_2(t) = Im\, Z(t)$ $\quad$ X_1, X_2 sol. p.t. $X' = AX$

X_1, X_2 sol. linearm. ind.

$\lambda = \alpha + i\beta$ $\quad$ $e^{\lambda t} = e^{(\alpha + i\beta)t} = e^{\alpha t + i\beta t} = e^{\alpha t} \cdot e^{i\beta t}$

Eulero $\quad$ $e^{it} = \cos t + i \sin t$

$e^{i\beta t} = \cos \beta t + i \sin \beta t$.

$e^{\lambda t} = e^{\alpha t}(\cos \beta t + i \sin \beta t)$

$v = v_1 + i v_2 \Rightarrow Z(t) = e^{\lambda t} v$

$= e^{\alpha t}(\cos \beta t + i \sin \beta t)(v_1 + i v_2)$

$$\Rightarrow \begin{cases} X_1(t) = e^{\alpha t}(\cos \beta t\, v_1 - \sin \beta t\, v_2) \\ X_2(t) = e^{\alpha t}(\sin \beta t\, v_1 + \cos \beta t\, v_2) \end{cases}$$

$Z : \mathbb{R} \longrightarrow \mathbb{C}^2$

X_1, X_2 reali sol. reali sist. di ec. diff. $X' = AX$

Z sol. complexă $\quad Z'(t) = AZ(t)$

$Z(t) = X_1(t) + iX_2(t)$

$\Rightarrow X_1'(t) + iX_2'(t) = A(X_1(t) + iX_2(t))$

$\Rightarrow X_1'(t) = AX_1(t)$

$X_2'(t) = AX_2(t)$

$\Rightarrow X_1, X_2 \in S_0$ sp. vectorial al sol. sist. de ec. dif. $X' = AX$

X_1, X_2 liniar indep. $\Leftrightarrow X_1(t_0), X_2(t_0) \in \mathbb{R}^2$ liniar indep. pt $t_0 \in \mathbb{R}$

Fie $t_0 = 0 \Rightarrow X_1(0) = v_1$, $X_2(0) = v_2$

v_1, v_2 liniar indep. $\Leftrightarrow \alpha_1 v_1 + \alpha_2 v_2 = 0 \Rightarrow \alpha_1 = \alpha_2 = 0$
$\alpha_1, \alpha_2 \in \mathbb{R}$

$\begin{cases} v = v_1 + iv_2 \\ \bar{v} = v_1 - iv_2 \end{cases}$

$v_1 = \frac{v+\bar{v}}{2}$, $v_2 = \frac{v-\bar{v}}{2i}$

$\alpha_1 v_1 + \alpha_2 v_2 = 0$

$\alpha_1 \left(\frac{v+\bar{v}}{2}\right) + \alpha_2 \left(\frac{v-\bar{v}}{2i}\right) = 0$

$\left(\frac{\alpha_1}{2} + \frac{\alpha_2}{2i}\right) v + \left(\frac{\alpha_1}{2} - \frac{\alpha_2}{2i}\right) \bar{v} = 0 \quad / \Leftrightarrow$

$v \in V_\lambda$, $\bar{v} \in V_{\bar{\lambda}}$ liniar indep.

$\Rightarrow \frac{\alpha_1}{2} + \frac{\alpha_2}{2i} = 0$, $\frac{\alpha_1}{2} - \frac{\alpha_2}{2i} = 0 \Rightarrow \alpha_1 = \alpha_2 = 0 \Rightarrow v_1, v_2$ liniar indep.

$\Rightarrow X_1, X_2$ liniar indep.

$\Rightarrow$ un sistem fundamental de soluții pt sist. de ec. dif. $X' = AX$ este

$X_1(t) = e^{\alpha t}(\cos\beta t \cdot v_1 - \sin\beta t \cdot v_2)$

$X_2(t) = e^{\alpha t}(\sin\beta t \cdot v_1 + \cos\beta t \cdot v_2)$

$\Rightarrow$ sol. generală $X(t) = C_1 X_1(t) + C_2 X_2(t)$, $C_1, C_2 \in \mathbb{R}$

exemplu: ecosistem — 2 specii $\begin{cases} x(t) & \text{prada} \\ y(t) & \text{prădător} \end{cases}$ (Lotka – Volterra)

prădă / prădător

$$\begin{cases} x' = 2x - 3y & x(0) = 5 & x(t) = ? \\ y' = 3x + 2y & y(0) = 4 & y(t) = ? \end{cases}$$

$$A = \begin{pmatrix} 2 & -3 \\ 3 & +2 \end{pmatrix} \qquad P_A(x) = x^2 - 4x + 13 = (x-2)^2 + 9 \quad \begin{cases} \lambda = 2+3i \\ \bar{\lambda} = 2-3i \end{cases}$$

$\lambda = 2 + 3i$ $\qquad V_\lambda$: $\begin{cases} 2x - 3y = (2+3i)x \\ 3x + 2y = (2+3i)y \end{cases} \Rightarrow y = -ix$

$$V_\lambda = \left\{ \begin{pmatrix} x \\ -ix \end{pmatrix} / x \in \mathbb{C} \right\} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ -i \end{pmatrix}}_{v} \right\}$$

$$Z(t) = e^{\lambda t} v = e^{(2+3i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} = e^{2t}(\cos 3t + i \sin 3t) \begin{pmatrix} 1 \\ -i \end{pmatrix} =$$

$$= e^{2t} \begin{pmatrix} \cos 3t + i \sin 3t \\ \sin 3t - i \cos 3t \end{pmatrix} \Rightarrow X_1(t) = e^{2t} \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix}$$ Sist. fundam. de soluții

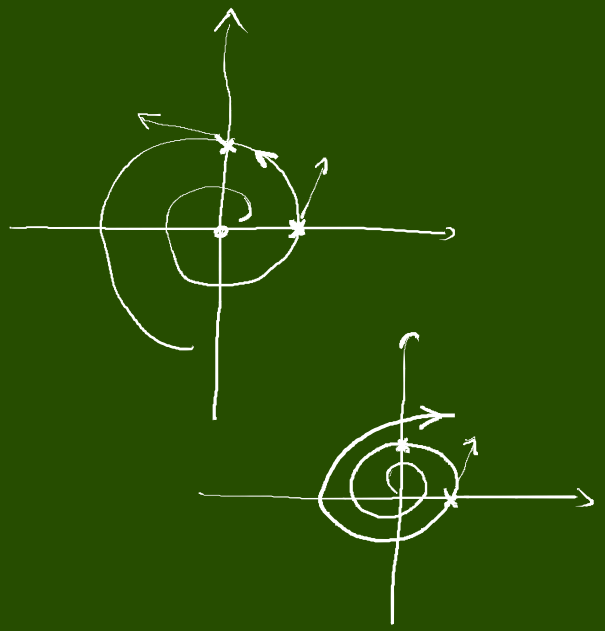
$$X_2(t) = e^{2t} \begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix}$$

$\Rightarrow$ sol. generale $X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 X_1(t) + C_2 X_2(t) =$

$= C_1 e^{2t} \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + C_2 e^{2t} \begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix} \quad C_1, C_2 \in \mathbb{R}$

periodo principale $\frac{2\pi}{3}$

Ritratto di fase



$$\begin{cases} x(t) = e^{2t}(C_1 \cos 3t + C_2 \sin 3t) \\ y(t) = e^{2t}(C_1 \sin 3t - C_2 \cos 3t) \end{cases}$$

$$X(t) = e^{2t} \underbrace{\begin{pmatrix} \cos 3t & \sin 3t \\ \sin 3t & -\cos 3t \end{pmatrix}}_{\text{matrice ortogonale}} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

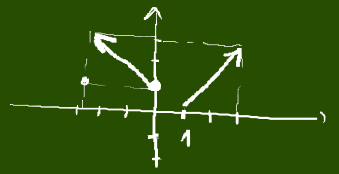
$$\sqrt{x^2+y^2} = e^{2t}\sqrt{C_1^2 + C_2^2}$$

$$x^2 + y^2 = \underbrace{(C_1^2 + C_2^2)}_{R^2} e^{4t}$$ spirale esponenziale

$\lim_{t \to \infty} (x^2+y^2) = \infty \quad (C_1 \neq 0, C_2 \neq 0)$

$$\begin{cases} x' = 2x - 3y \\ y' = 3x + 2y \end{cases}$$

$x = 1,\ y = 0 \Rightarrow x' = 2,\ y' = 3$



$\Rightarrow$ sens trigonometric

$x = 0,\ y = 1 \Rightarrow x' = -3,\ y' = 2$

$x(0) = 5 \qquad y(0) = 4$

$$\begin{cases} x(t) = e^{2t}(C_1 \cos 3t + C_2 \sin 3t) \\ y(t) = e^{2t}(C_1 \sin 3t - C_2 \cos 3t) \end{cases} \Rightarrow \begin{matrix} C_1 = 5 \\ -C_2 = 4 \end{matrix}$$

$$\begin{cases} x(t) = e^{2t}(5\cos 3t - 4\sin 3t) \\ y(t) = e^{2t}(5\sin 3t + 4\cos 3t) \end{cases} \quad t \geq 0$$

$\frac{5\cos 3t - 4\sin 3t}{5\sin 3t + 4\cos 3t} = 0 \Rightarrow t_1$, $t_1 \in (0, \frac{\pi}{2})$

$5\cos 3t - 4\sin 3t = 0 \Rightarrow \operatorname{tg} 3t = \frac{5}{4}$

$5\sin 3t + 4\cos 3t = 0 \Rightarrow t_2$

$\Rightarrow \operatorname{tg} 3t = -\frac{4}{5}$

$\Rightarrow t_2 \in (\frac{\pi}{2}, \pi)$

$$\begin{cases} x' = 2x - 3y \\ y' = 3x + 2y + 1 \end{cases}$$

$\longleftrightarrow$ sistem de ec. dif. liniare neomogene

$X' = AX + b(t)$ $\quad$ t variabila

$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ fct. necc. $\quad$ $x_1 = x_1(t)$ ... $x_n = x_n(t)$

$A \in \mathcal{M}_n(\mathbb{R})$

$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}^n$ fct. cont.

Thm. Problema Cauchy $\begin{cases} X' = AX + b(t) \\ X(t_0) = X_0 \end{cases}$ are sol. unică $\forall\, t_0 \in I$, $\forall\, X_0 \in \mathbb{R}^n$

Prop. 1 $S_o = \{X : \mathbb{R} \longrightarrow \mathbb{R}^n \mid X \text{ deriv.}, X' = AX\}$ sp. vect. al sol. sist. de ec. dif. omogene

$S_{no} = \{X : I \longrightarrow \mathbb{R}^n \mid X \text{ deriv.}, X' = AX + b(t)\}$ multimea sol. sist. de ec. dif. neomogene

Fie $X_p \in S_{no}$ o sol. particulară. Atunci:

$X \in S_{n.o.} \iff X - X_p \in S_o$

$X' = AX + b(t)$

$X_p' = AX_p + b(t)$

$(X - X_p)' = A(X - X_p)$

Concl. X_o sol. gen. a ec. omogene

X_p sol. part. a ec. neomogene

$\Rightarrow X = X_o + X_p$ sol. gen. a ec. neomogene.

Exemplu.

$$\begin{cases} x' = 2x - 3y + 0 \\ y' = 3x + 2y + 1 \end{cases}$$

$$X_o(t) = C_1 e^{2t}\begin{pmatrix}\cos 3t \\ \sin 3t\end{pmatrix} + C_2 e^{2t}\begin{pmatrix}\sin 3t \\ -\cos 3t\end{pmatrix}$$

$$X_p(t) = \begin{pmatrix}-3/13 \\ -2/13\end{pmatrix}$$

$$\begin{cases} 0 = 2a - 3b & /\cdot 2 \\ -1 = 3a + 2b & /3 \end{cases}$$

$-3 = 13a \qquad a = -\frac{3}{13} \qquad b = \frac{2a}{3} = -\frac{2}{13}$

$$\Rightarrow X(t) = X_o(t) + X_p(t) = C_1 e^{2t}\begin{pmatrix}\cos 3t \\ \sin 3t\end{pmatrix} + C_2 e^{2t}\begin{pmatrix}\sin 3t \\ -\cos 3t\end{pmatrix} + \begin{pmatrix}-3/13 \\ -2/13\end{pmatrix}$$

Prop. 2 Fie $X_1(t), \dots, X_n(t) \in S_0$ un sistem fundamental de sol. pt. sist. de ec. dif. omogen $X' = AX$.

(Sol. gen. a sist. omogen este $X_0(t) = C_1 X_1(t) + \dots + C_n X_n(t)$)

$\Rightarrow$ Sol. gen. a sist. neomogen va fi $X(t) = C_1(t) X_1(t) + \dots + C_n(t) X_n(t)$

unde $C_1(t), \dots, C_n(t)$ verifică $C_1'(t) X_1(t) + \dots + C_n'(t) X_n(t) = b(t)$

(metoda variației constantelor)

$X(t) \in S_{n0} \iff X' = AX + b(t) \iff$

$C_1'(t) X_1(t) + C_1(t) X_1'(t) + \dots + C_n'(t) X_n(t) + C_n(t) X_n'(t) =$

$A C_1(t) X_1(t) + \dots + A C_n(t) X_n(t) + b(t)$

$X_1 \dots X_n \in S_0 \Rightarrow X_i' = A X_i \quad i = \overline{1, n}$

$\Rightarrow C_1'(t) X_1(t) + \dots + C_n'(t) X_n(t) = b(t)$

Sistem n ec. n necun $C_1'(t), \dots C_n'(t)$ - sol. unică.

$$\begin{cases} x' = 2x - 3y + 0 \\ y' = 3x + 2y + 1 \end{cases}$$

$$X_0(t) = C_1 e^{2t}\begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + C_2 e^{2t}\begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix}$$

$$\begin{cases} x_0(t) = C_1 e^{2t}\cos 3t + C_2 e^{2t}\sin 3t \\ y_0(t) = C_1 e^{2t}\sin 3t - C_2 e^{2t}\cos 3t \end{cases}$$

Cauf. sol. sist. neomogen de forma

$\begin{cases} x(t) = C_1(t) e^{2t}\cos 3t + C_2(t) e^{2t}\sin 3t \\ y(t) = C_1(t) e^{2t}\sin 3t - C_2(t) e^{2t}\cos 3t \end{cases}$ unde $C_1(t)$, $C_2(t)$ verifică nel:

$$\begin{cases} C_1'(t) e^{2t}\cos 3t + C_2'(t) e^{2t}\sin 3t = 0 \quad / \cos 3t \\ C_1'(t) e^{2t}\sin 3t - C_2'(t) e^{2t}\cos 3t = 1 \quad / \sin 3t \end{cases}$$

$$C_1'(t) e^{2t} = \sin 3t \Rightarrow C_1'(t) = e^{-2t}\sin 3t$$

$$C_2'(t) e^{2t} = -\cos 3t \Rightarrow C_2'(t) = -e^{-2t}\cos 3t.$$

$$C_1(t) = \int e^{-2t}\sin 3t\, dt = \frac{e^{-2t}}{-2}\sin 3t - \int \frac{e^{-2t}}{-2}\cos 3t \cdot 3\, dt =$$

$$= -\frac{1}{2}e^{-2t}\sin 3t + \frac{3}{2}\int e^{-2t}\cos 3t\,dt =$$

$$= -\frac{1}{2}e^{-2t}\sin 3t + \frac{3}{2}\left(\frac{e^{-2t}}{-2}\cos 3t + \int \frac{e^{-2t}}{-2}\,3\sin 3t\,dt\right)$$

$$= -\frac{1}{2}e^{-2t}\sin 3t - \frac{3}{4}e^{-2t}\cos 3t - \frac{9}{4}C_1(t)$$

$$C_1(t) = \frac{4}{13}e^{-2t}\left(-\frac{1}{2}\sin 3t - \frac{3}{4}\cos 3t\right) + D_1$$

$$C_2(t) = -\int e^{-2t}\cos 3t\,dt = -\frac{2}{3}\left(C_1(t) + \frac{1}{2}e^{-2t}\sin 3t\right) =$$

$$= -\frac{2}{3}e^{-2t}\left(\left(-\frac{2}{13}+\frac{1}{2}\right)\sin 3t - \frac{3}{13}\cos 3t\right)$$

$$= e^{-2t}\left(-\frac{1}{13}\sin 3t + \frac{2}{13}\cos 3t\right) + D_2$$

$$x(t) = C_1(t)e^{+2t}\cos 3t + C_2 e^{-2t}\sin 3t$$

$$= \frac{4}{13}\left(-\frac{1}{2}\sin 3t - \frac{3}{4}\cos 3t\right)\cos 3t + D_1 e^{-2t}\cos 3t +$$

$$\left(-\frac{3}{13}\sin 3t + \frac{2}{13}\cancel{\cos 3t}\right)\sin 3t + D_2 e^{2t}\sin 3t$$

$$= \underbrace{D_1 e^{2t}\cos 3t + D_2 e^{2t}\sin 3t}_{x_0(t)} + \underbrace{\left(-\frac{3}{13}\right)}_{x_p(t)}$$

$$y(t) = C_1(t)e^{2t}\sin 3t - C_2(t)e^{2t}\cos 3t = \dots = y_0(t) + y_p(t)$$

$$\begin{cases} x' = 2x - 3y & x(0) = 5 \\ y' = 3x + 2y + 1 & y(0) = 4 \end{cases}$$

$$\Rightarrow \begin{cases} D_1 - \frac{3}{13} = 5 \\ -D_2 - \frac{2}{13} = 4 \end{cases} \Rightarrow D_1 = \frac{68}{13}$$

$$D_2 = -\frac{54}{13}$$

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esercizio

$$\begin{cases} x' = x + 2y + 3e^{-t} \\ y' = 2x + y - e^{-t} \end{cases}$$

$$x_p(t) = ae^{-t}e^{2t}$$

$$y_p(t) = be^{-t}$$

$$-ae^{-t} = ae^{-t} + 2be^{-t} + 3e^{-t}$$

$$-be^{-t} = 2ae^{-t} + be^{-t} - e^{-t}$$

$$\begin{cases} 2a + 2b + 3 = 0 \\ 2a + 2b - 1 = 0 \end{cases}$$

system in complex form

($\lambda = -1$ val. par. per $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$)

$e^{\lambda t} v$